Global SL(2,C) recoupling and a normalization-independent recoil test

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Abstract

We fix principal-series conventions, shadow equivalence and the spectral measure needed for a proposed $G = 0$ recoil dictionary. A spectral rectangle removes arbitrary local channel normalizations. For one explicit topology-preserving scalar identification, its value for the known global conformal $6j$ differs from both the recoil function H and the paired graph $Z_{\rm rec}$. High precision and an independent Euler representation support the mismatch; no interval certificate is claimed. A separate positivity argument rules out a continuous nonnegative kernel as a complete scalar-only unitary transform. Neither result excludes nonlocal dictionaries or constructs finite-gravity de Sitter dynamics.

1 Principal series and the meaning of a barred weight

The reference harmonic analysis is on a Euclidean two-dimensional boundary. We set the de Sitter radius to one when comparing labels; no physical gravitational inner product is inferred from this choice. For bosonic single-valued representations,

$$\Delta = 1 + i\nu, \quad J \in \mathbb{Z}, \qquad h = \frac{1 + J + i\nu}{2}, \quad \bar{h} = \frac{1 - J + i\nu}{2}. \tag{1}$$

Here $\bar{h}$ is an antiholomorphic label, *not* complex conjugation. Unitarity obeys $h^* + \bar{h} = 1$; a scalar has $h = \bar{h} = (1 + i\nu)/2$. The single-representation space has inner product $\int_{\mathbb{C}} \mathrm{d}^2 z\, f(z)^* g(z)$ [2]. This is not yet the Hilbert space of the paired Virasoro theory.

Use equation (5) of Derkachov–Spiridonov (DS) [1] to fix their integer convention:

$$a = m/2 + i\sigma, \quad \bar{a} = -m/2 + i\sigma, \quad h = (1 - a)/2, \quad \bar{h} = (1 - \bar{a})/2. \tag{2}$$

Thus $J = -m/2$ and $\nu = -\sigma$. We restrict to even m. Distinct scalar external legs still permit every integer internal spin. Identical-field symmetry restrictions require a separate exchange-parity analysis.

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2 Reflection, measure and the two-factor obstruction

DS equations (22), (28)–(30) contain a direct $\delta_R(a-a')$ contribution and a reflected $\delta_R(a+a')$ contribution multiplied by a shadow kernel. In their CG convention,

$$\rho_{\rm DS}(a) = -\frac{a\bar{a}}{4\pi^4} = \frac{m^2/4+\sigma^2}{4\pi^4}, \qquad \int \frac{\rho_{\rm DS}(a)}{2} D_R a = \frac{1}{2}\sum_{m\in 2\mathbb{Z}} \int_{\mathbb{R}} {\rm d}\sigma\, \rho_{\rm DS}(a). \tag{3}$$

The factor $1/2$ accounts for reflected equivalent representations. Choosing a fundamental domain also changes basis bookkeeping; one cannot simply discard the reflection term. The Racah transform in DS (32) uses this measure. Their Barnes formula (45) contains an integer sum and an integral. Scalar external legs alone do not remove the sum.

In the distinct KKS normalization, equations (73)–(74) give

$$\mathcal{N}(\Delta,0) = \frac{\pi^2\Gamma(1-\Delta)\Gamma(\Delta-1)}{\Gamma(\Delta)\Gamma(2-\Delta)} = \frac{\pi^2}{\nu^2}, \qquad \mu_{\rm KKS} = \frac{1}{4\mathcal{N}\ {\rm vol}\, SO(2)}. \tag{4}$$

Their partial-wave norm also includes two three-point pairings, equation (39). Raw DS density, KKS measure and a four-point inverse norm are not interchangeable.

For $\widehat{f}(p) = \int {\rm d}^2x\, e^{-ip\cdot x} f(x)$, the distributionally continued scalar shadow kernel $|x-y|^{-2(2-\Delta)}$ has multiplier

$$\widehat{S}_\Delta(p) = \pi\, 2^{2i\nu} \frac{\Gamma(i\nu)}{\Gamma(1-i\nu)} |p|^{-2i\nu}. \tag{5}$$

Its modulus is π/ν for $\nu > 0$. Multiplication by ν/π defines a unitary Fourier multiplier on $L^2(\mathbb{C})$; $p=0$ has measure zero. The endpoint $\nu = 0$ is excluded.

The algebraic relation $z(M) = M^2 e^{-2\pi M} = 4\pi^4 e^{-2\pi M}\rho_{\rm DS}(iM)$ does not derive the physical observer measure. The exponential weight is additional state or preparation information.

The common-angle, fixed-mass limit of the two CLP factors [4], checked in our preceding note, is

$$h_+ = (1-i\mu)/2, \qquad h_- = (1+i\mu)/2. \tag{6}$$

Identifying these directly with $(h,\bar{h})$ gives $J = -i\mu$, incompatible with (1) for generic real μ. Formally replacing a factor by a dual label is not a constructed physical shadow transform. We instead test a separately defined global scalar dictionary below.

3 The global matrix element and the candidate map

The LPRS conformal $6j$ [3] is defined by the bilinear overlap of a shadow partial wave and a partial wave, their (3.5). It is not itself a unit-normalized matrix. Changing channel norms only produces row and column factors.

All weights below are scalar: $h_i = \bar{h}_i$, $h = \bar{h}$ and $h' = \bar{h}'$. Set $h_{ij} = h_i - h_j$, $p = h_2 + h_3$. LPRS (3.17), (3.35)–(3.37) give

$$K_q^{ab} = \pi \frac{\Gamma(2q-1)}{\Gamma(2-2q)} \frac{\Gamma(1-q-h_{ab})\Gamma(1-q+h_{ab})}{\Gamma(q-h_{ab})\Gamma(q+h_{ab})}, \tag{7}$$

$$\begin{aligned}\Omega_{x,y} = &\int_0^1 \mathrm{d}u\, u^{x+p+h_{13}-2}(1-u)^{y-p} \\ &\times {}_2F_1(x+h_{12}, x-h_{34}; 2x; u)\,{}_2F_1(y-h_{32}, y+h_{14}; 2y; 1-u),\end{aligned} \tag{8}$$

$$\begin{aligned}B(h,q) = &\frac{\Gamma(h+h_{12})\Gamma(h-h_{12})\Gamma(1-h+h_{43})\Gamma(1-h+h_{34})}{4\Gamma(2h)\Gamma(2-2h)} \\ &\times \sin\pi(q-h_1-h_4)\sin\pi(q-h_2-h_3)\Omega_{h,q}\Omega_{1-h,q},\end{aligned} \tag{9}$$

$$\mathcal{J}_2(h,h') = K^{14}_{1-h'}B(h,h') + K^{32}_{h'}B(h,1-h'). \tag{10}$$

The implementation retains both shadow terms in (10) and both ${}_4F_3$ terms in each LPRS (3.38) expression for Ω. Equation (8) is an independent check. For our samples its endpoint real powers are $-1/2$, with at most logarithmic factors, so the integral converges.

The frozen recoil graph has vertices $V = \{123, 156, 246, 345\}$. Its opposite internal edges are $\mu_3 = s$ and $\mu_6 = t$. The explicit assignment is

Global role	Recoil labels	Representation
External $1,2,3,4$	$\mu_1, \mu_2, \mu_4, \mu_5$	$\Delta_e = 1 + i\mu_e,\ J_e = 0$
s channel	$\mu_3 = s$	$\Delta_s = 1 + is,\ J_s = 0$
t channel	$\mu_6 = t$	$\Delta_t = 1 + it,\ J_t = 0$

The s vertices join $(12), (34)$; the t vertices join $(14), (23)$, preserving all four recoil vertices. These are representation labels, not newly identified bare masses.

The hypothesis is $K(s,t) = A(s)B(t)\mathcal{J}_2(s,t)$ with arbitrary nonzero complex A, B, for $K = H$ or

$$Z_{\mathrm{rec}} = \pi^6 \prod_{e=1}^{6} \frac{\mu_e}{\sinh \pi\mu_e} |H|^2 \quad (\mu_e > 0). \tag{11}$$

Every local vertex normalization depends on only s or t with external labels fixed. These factors and absorbed channel measures are included; arbitrary joint factors are not.

4 A normalization-independent numerical test

For four nonzero matrix elements, define

$$\mathfrak{C}[K] = \frac{K(s_1,t_1)K(s_2,t_2)}{K(s_1,t_2)K(s_2,t_1)}. \tag{12}$$

Substitution proves $\mathfrak{C}[ABK] = \mathfrak{C}[K]$. Use

$$\begin{gathered}(\mu_1,\mu_2,\mu_4,\mu_5) = (.4,.7,.93,.6),\\ (s_1,s_2) = (.65,.85), \qquad (t_1,t_2) = (.57,.81).\end{gathered} \tag{13}$$

All vertices satisfy strict triangle inequalities and the left and right recoil residue families are nondegenerate. At 60 decimal digits,

$$\mathfrak{C}[\mathcal{J}_2] = 0.97254633802865920965 + O(10^{-60})i, \tag{14}$$

$$\mathfrak{C}[H] = 1.00800263623569531689 - 0.03406416860860421339i, \tag{15}$$

$$\mathfrak{C}[Z_{\mathrm{rec}}] = 1.01722968224110691421. \tag{16}$$

The relative discrepancies $|\mathfrak{C}[K]/\mathfrak{C}[\mathcal{J}_2] - 1|$ are 0.05055620141488165 and 0.04594469431967669. Equation (11) implies $\mathfrak{C}[Z_{\mathrm{rec}}] = |\mathfrak{C}[H]|^2$: the six leg factors cancel.

These differences survive local leg, vertex and channel renormalizations under the stated map. They are *finite-point numerical counterexample evidence*, not an interval-certified inequality. They do not exclude every physical dictionary or the CLP fixed-topology proposal.

Independent diagnostic	Relative discrepancy
40 to 60 digits, points and rectangles (maximum)	2.34×10^{-39}
First complete $\mathcal{J}_2$: direct Euler vs. hypergeometric	4.36×10^{-33}
Four Euler Ω checks (maximum)	5.75×10^{-33}
Three shadow-representative checks (maximum)	1.52×10^{-60}

The maximum recoil residue cancellation ratio is about 220.22; the two-shadow ratio in $\mathcal{J}_2$ is at most 1.108. The shadow checks separately flip s, t, or external μ_1; they do not exhaust all permutations or transformations.

There are 44 passed diagnostics: eight left/right recoil checks, 12 pointwise precision comparisons, three rectangle refinements, one Z/H identity, five direct Euler comparisons, three shadow checks and 12 shadow-norm/multiplier checks. Python 3.14.5 and mpmath 1.3.0 were used. Full complex values, parameters, dependency versions and source hashes are retained.

An initial parameter choice met merged residue families and raised a Gamma-pole exception in the separated formula. The initial driver and failure record were saved. We moved to the generic rectangle without changing frozen recoil code or weakening tolerances. This formula-domain failure is not a demonstrated singularity of H.

5 An exact obstruction and the next physical task

Proposition 1. *Let two continuous label spaces carry locally finite positive Borel measures with full support on open intervals. A continuous, everywhere nonnegative kernel cannot define a complete unitary integral transform between their L^2 spaces.*

Proof. A nonzero kernel is strictly positive on some open rectangle. Choose two nonnegative, nonzero input test functions with disjoint compact supports within its input interval. Their inner product vanishes, while both outputs are strictly positive on the output interval. Nonnegativity elsewhere prevents cancellation, so their output inner product is positive. If either output fails to lie in L^2, unitarity already fails. The zero kernel is not unitary. □

Thus the nonnegative real-mass Z_{rec} cannot be a *complete scalar-only* unitary recoupling kernel repaired solely by positive weights. Separable basis phases can be removed by unitary rephasings. This does not apply to complex H, a scalar subblock of a spin-summed unitary transform, a Euclidean amplitude, a probability kernel, or a physical correlator. Missing spin channels matter for orthogonality.

The next task should derive a physical insertion: start with $G = 0$ dS_3 Bunch–Davies positive-frequency modes, Klein–Gordon normalization and observer representations. Define the local-field matrix element and its invariant or group-averaged three-point integral; match its full three-label coefficient and reflection factors to the DS CG kernel. Covariance alone leaves coupling coefficients and does not fix dynamics. This comparison can test whether the undetermined two-point normalization n is fixed.

A restriction to spin zero must specify actual projectors and the resulting incompleteness. Retain the independent free-probe, large-observer-mass benchmark

$$g_\nu(0)g_{\nu'}(0) = \frac{\nu\nu'}{16\pi^2 \sinh \pi\nu \, \sinh \pi\nu'}. \tag{17}$$

This is a disconnected antipodal product, not a zero-energy graph value. Physical finite-κ sewing, finite-G dynamics and complete de Sitter holography remain open. This note narrows a candidate dictionary; it does not supply a new holographic dual.

References

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[3] J. Liu, E. Perlmutter, V. Rosenhaus and D. Simmons-Duffin, *d-dimensional SYK, AdS Loops, and 6j Symbols*, arXiv:1808.00612v3.

[4] J. Chandra, D. Liska and B. Post, *3D de Sitter from complex Virasoro TQFT: Observers and knots*, arXiv:2610.30247v1.